

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've spent 10 years writing software, with the last 3 focused on technical leadership. I own API contracts, write RFCs, and sit on our Architecture Steering Committee, but I still write code daily. I built a developer platform that cut environment provisioning from weeks to hours. I run production systems on AWS and Kubernetes with 99.9% uptime. I've worked on payment integrations and billing systems. My career spans Python, TypeScript, Java, C#, and Gosu across insurance, medical imaging, and defense. I pick up new stacks quickly and care about simple solutions that make teams more effective.

Work Experience

Alberta Motor Association

SOFTWARE ARCHITECT

Edmonton, Alberta

Jan 2023 - Present

- Architected backend services for automated phone payment system. Coordinated across 2 internal teams and 2 external partners (Replicant chatbot, Stripe). Launched Dec 2025. (Python, AWS)
- Own 4 REST APIs serving 5 internal teams and external partners. Design contracts, maintain OpenAPI specs, approve new endpoints. (Python, TypeScript, REST, OpenAPI)
- Architected infrastructure for pay-as-you-go insurance product across multiple AWS accounts. 2.8M events processed, 76K invoices, 99.9% uptime. Won 2024 Guidewire Innovation Award. (Python, AWS Lambda, TypeScript, Terraform)
- Built self-service developer platform: environment provisioning dropped from weeks to 2 hours. Adopted by 15 engineers across 3 teams, created 120+ environments. (Python, Terraform, AWS, Docker)
- Led modernization of document generation from legacy C# to containerized microservices on EKS. Led 2-3 engineers, added Datadog observability. (Kubernetes, Docker, Datadog, Helm)
- Mentor engineers on system design and debugging. Author RFCs, member of Architecture Steering Committee.

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built and maintained backend services for SaaS medical imaging platform. Services ran AI algorithms and returned results to the platform. (Node.js, TypeScript, AWS)
- Improved AI algorithm processing turnaround using Redis caching and AWS infrastructure tuning.
- Monitored production via Elasticsearch and Kibana. Caught issues before customer impact.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Led 5-6 person team on Guidewire insurance platform. Delivered projects on time with no critical defects post-launch. Code reviews, mentored junior developers, collaborated with product stakeholders.
- Planned and implemented online payment integrations with third-party APIs.
- Built automated test framework that reduced defects by 30% on subsequent releases. Framework adopted org-wide, still in use 4+ years later. (Java/Gosu, JUnit)
- Led Git migration for 12-15 developers. Created Docker image for local dev environments.

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD. (C#)

Lockheed Martin Canada

SOFTWARE ENGINEER ASSOCIATE

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining real-time data streams from multiple sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system for capturing and replaying test scenarios, improving debugging and test coverage.

The King's University

SOFTWARE DEVELOPER/RESEARCH ASSISTANT

Edmonton, Alberta

May 2014 - Sep 2016

- Built web application for spatial/temporal data visualization. (JavaScript, PostgreSQL, PHP)

Education

The Open University
MSC. COMPUTING (SOFTWARE ENGINEERING)
Currently pursuing Master's in Computing at The Open University.

Milton Keynes, England (Remote)
May 2024 - Ongoing

The King's University
B.S. IN COMPUTER SCIENCE

Edmonton, Alberta
Sep 2012 - Dec 2016

Skills

Distributed Systems

Build backend systems on AWS with 99.9% uptime. Event-driven pipelines, microservices on Kubernetes, multi-account architectures. Focus on reliability and clean abstractions.

API Design & Ownership

Own 4 REST APIs serving internal teams and external partners. Design contracts, maintain OpenAPI specs, run code reviews. Experience with billing systems and payment integrations.

Technical Leadership & Mentoring

Lead architecture decisions while staying hands-on. Write design docs (RFCs), run code reviews, mentor engineers through pairing. Member of Architecture Steering Committee.

Fast Learner

Pick up new languages and frameworks quickly. Career spans Python, TypeScript, Java, C#, Gosu across insurance, medical imaging, and defense domains. Comfortable ramping on unfamiliar stacks.

Technologies

Languages Python, TypeScript, Java, C#, Bash

Web & APIs REST APIs, OpenAPI, Node.js

Cloud & Infrastructure AWS (Lambda, EKS, EC2, RDS, S3, SQS), Kubernetes, Docker, Terraform

Databases PostgreSQL, Aurora/RDS, Redis

Observability Datadog, Elasticsearch, Kibana